

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Zoology**(Ancillary Course)****Paper: ZOB-203A — Ancillary Biology (Animal Biology)**

Time: Three hours

Full Marks: 70

Note: Answer five questions including Question 1 which is compulsory. Draw diagrams wherever necessary. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

(a) Select and write the best option in the following: [8 × $\frac{1}{2}$ = 4]

(i) Which of the following is not in the correct order of hierarchy for classification?

- (1) Phylum → Class → Family → Order
- (2) Phylum → Class → Order → Genus
- (3) Class → Order → Genus → Species
- (4) Class → Order → Family → Genus

(ii) The theory of Natural Selection was given by:

- (1) Lamarck
- (2) Darwin
- (3) Wallace
- (4) Crick

(iii) The alleles are:

- (1) the genes present on same chromosomes
- (2) the two genes on two different chromosomes
- (3) alternate form of same gene
- (4) a pair of different genes

(iv) Which Law of Mendel is revealed by a mono-hybrid cross?

- (1) Law of dominance
- (2) Law of independent assortment
- (3) Law of heterozygosity
- (4) Law of segregation

(v) Maltotriose is an example of:

- (1) monosaccharide
- (2) polysaccharide
- (3) disaccharide
- (4) oligosaccharide
- (vi) Testosterone is secreted by:
- (1) granulosa cells
- (2) leydig cells
- (3) sertoli cells
- (4) sperms
- (vii) Which one of the following is a principal digestive gland?
- (1) Bladder
- (2) Liver
- (3) Kidney
- (4) Stomach
- (viii) Diabetes mellitus is a disease associated with metabolism of:
- (1) carbohydrate
- (2) lipid
- (3) protein
- (4) nucleic acid

(b) Identify the best match for terms in Column I with those in Column II: $[8 \times \frac{1}{2} = 4]$

Column I	Column II
(i) Homology	(1) α and β forms of ring structure of sugar
(ii) <i>Homo erectus</i>	(2) differ in configuration around specific carbon
(iii) Golgi complex	(3) transient, dissociable binding to enzyme or to substrate
(iv) Down syndrome	(4) Same origin but may differ in function
(v) Inspiration	(5) secretion
(vi) Epimers	(6) closest to modern man
(vii) Anomers	(7) chromosomal abnormality
(viii) Cofactors	(8) external intercostal muscles
	(9) protein synthesis

(c) Write whether the following statements are True or False: $[8 \times \frac{1}{2} = 4]$

- (i) Classification of living organisms reflects their evolutionary relationship.
- (ii) Cell membrane is mainly made up of phosphoprotein bilayer.

- (iii) Chromosome number becomes half in first meiotic division.
 - (iv) Reverse translation is also a way of genetic information flow.
 - (v) Macrophages produce antibodies.
 - (vi) Prosthetic groups in enzyme are small non-protein molecules and metal ions that participate directly in catalysis.
 - (vii) Transferase catalyses geometric or structural changes within a single molecule.
 - (viii) The pH inside stomach is 8.9.
- (d) Define the following: [10 × 1 = 10]
- (i) Organic evolution
 - (ii) Eukaryote
 - (iii) Apoptosis
 - (iv) NK cells
 - (v) Cytokine
 - (vi) Binding energy
 - (vii) Endocrine gland
 - (viii) Residual volume of lung
 - (ix) Homology
 - (x) Hormone.

Question 2. Illustrate the following with labelled diagram: [6 × 2 = 12]

- (a) An eukaryotic cell
- (b) A nephron.

Question 3. Distinguish between: [6 × 2 = 12]

- (a) Mitosis and meiosis
- (b) Immune response mediated by T-cells and B-cells.

Question 4. Answer the following parts:

- (a) Give an example of Natural selection. Discuss it in brief. [6]
- (b) Describe the mechanism of enzyme action. [6]

Question 5. Answer the following parts:

- (a) Prepare a chart to show the major events that occur in different phases of meiosis. [6]
- (b) Illustrate the counter current mechanism in Kidney during urine formation. [6]

Question 6. Write short notes on any three of the following:

[4 × 3 = 12]

- (a) Human evolution
- (b) Cellular respiration
- (c) Peptide bond
- (d) Sickle cell anemia.

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